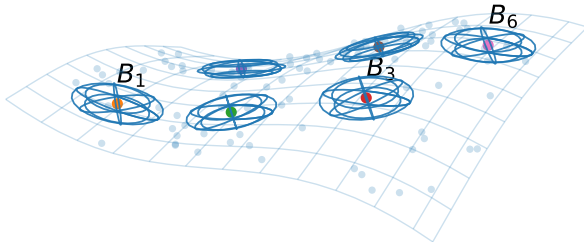


Normality Bubble Diffusion

Established methods and an MVTec AD study



Huynh Thanh Dat 15 September 2026

Methods → Our benchmarks → NBD → Comparison

One-class detection

Train on normal data

$$\mathcal{D}_N = \{x_i\}_{i=1}^n.$$

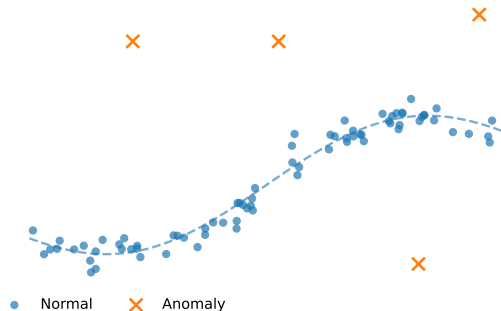
Score new observations

$$s(x) : \mathcal{X} \rightarrow \mathbb{R}.$$

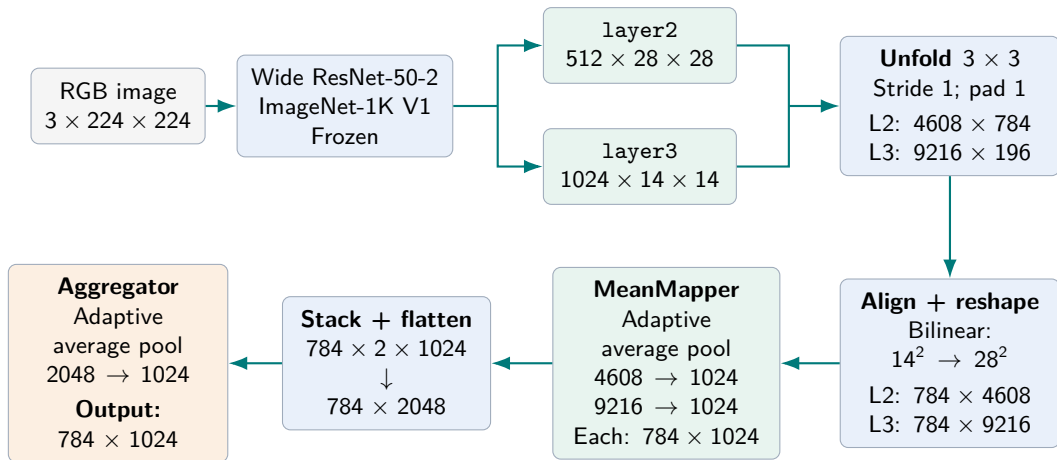
A larger score means more anomalous.

Model fitting: normal observations.

Test: normal + anomaly.



PatchCore: image-to-patch feature map



PatchCore: build memory, then score new images

Build normal memory

Collect normal patch descriptors.

Greedy coreset: keep **10%**.

One memory per category.

Get the score

Patch: distance to the nearest memory patch.

$$s_i = \min_{m \in \mathcal{M}} \|z_i - m\|_2.$$

Image: take the largest patch score, then apply neighborhood reweighting.

$$s^* = \max_i s_i,$$
$$s(x) = \left(1 - \frac{e^{\|m_{\text{test},*} - m^*\|_2}}{\sum_{m \in \mathcal{N}_b(m^*)} e^{\|m_{\text{test},*} - m\|_2}} \right) s^*.$$

Map: arrange patch scores, upsample, smooth (Gaussian blur).

PaDiM: a distributional view of patch anomalies

PatchCore / PatchScore

Keep normal patches in a memory bank.

New patch $\rightarrow$ nearest stored patch.

PaDiM

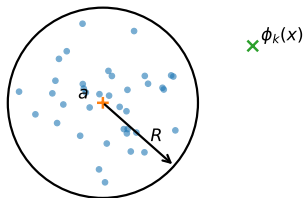
Fit one Gaussian per patch position.

New patch $\rightarrow$ distance from its Gaussian.

$$X_{ij} \sim \mathcal{N}(\mu_{ij}, \Sigma_{ij}), \quad s_{ij} = \sqrt{(x_{ij} - \mu_{ij})^T \Sigma_{ij}^{-1} (x_{ij} - \mu_{ij})}.$$

Defard et al., *PaDiM: A Patch Distribution Modeling Framework for Anomaly Detection and Localization*, ICPR 2020.

Kernel SVDD: enclose normal data in a sphere



Fit a compact boundary

The kernel maps inputs into a space where normal observations lie near one center.

$$\min_{a, R, \xi \geq 0} R^2 + \frac{1}{\nu n} \sum_i \xi_i, \quad \|\phi_k(x_i) - a\|^2 \leq R^2 + \xi_i.$$

$$s(x) = \|\phi_k(x) - a\|^2 - R^2.$$

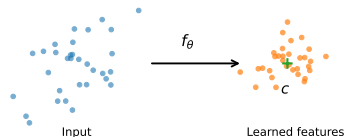
Image benchmark in Ruff et al. (2018)

Flatten images; fit PCA to retain 95% of normal training variance. Use the Gaussian kernel:

$$k(x, x') = \exp(-\gamma \|x - x'\|_2^2).$$

Search $\gamma \in \{2^{-10}, \dots, 2^{-1}\}$ and $\nu \in \{0.01, 0.1\}$.

Deep SVDD: learn a compact normal representation



Train the image encoder

An autoencoder initializes the CNN. Deep SVDD then brings normal image embeddings toward a fixed center c .

MNIST: LeNet, 8/4 channels, 32-D output.

CIFAR-10: LeNet, 32/64/128 channels, 128-D output.

Full reproduction schedule

MNIST: AE 150 $\rightarrow$ SVDD 150 epochs.

CIFAR-10: AE 350 $\rightarrow$ SVDD 150 epochs.

BASELINE METHODS

One-class objective

$$\min_{\theta} \frac{1}{n} \sum_i \|g_{\theta}(x_i) - c\|^2 + \frac{\lambda}{2} \|\theta\|^2.$$

Soft-boundary objective

$$\min_{\theta, R} R^2 + \frac{1}{\nu n} \sum_i [\|g_{\theta}(x_i) - c\|^2 - R^2]_+ + \frac{\lambda}{2} \|\theta\|^2.$$

Score a complete image

$$s(x) = \|g_{\theta}(x) - c\|^2.$$

For the soft boundary, subtract R^2 ; ranking is unchanged.

Deep SVDD: preventing collapse

The failure: one output for every input

A constant mapping makes normal and anomalous observations indistinguishable.

Trivial solution

$$g_{\theta}(x) \equiv c \implies s(x) = 0.$$

A trainable bias can produce this mapping:

$$g_{\theta}(x) = Wh(x) + b, \quad W = 0, \quad b = c.$$

Preventing collapse

Bias-free layers. Remove the direct constant-output shortcut.

Fixed nonzero center. Estimate c from normal outputs and keep it fixed during training.

ReLU / leaky ReLU. Avoid bounded activations with nonzero saturation.

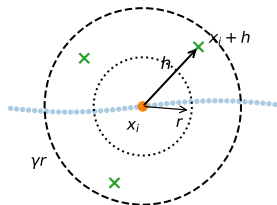
Initialization

Pretrain an autoencoder on normal images, then transfer its encoder weights.

DROCC: generate negatives around normal images

$$\min_{\theta} \lambda \|\theta\|^2 + \sum_i \left[\ell(g_{\theta}(x_i), +1) + \mu \max_{\tilde{x} \in \mathcal{N}_i(r)} \ell(g_{\theta}(\tilde{x}), -1) \right].$$

Search in the annulus



Generate a hard negative

1. Start from $\tilde{x} = x_i + h$ near a normal sample.
2. Make it harder by projected gradient ascent:

$$h \leftarrow \Pi_{[r, \gamma r]} \left(h + \eta \frac{\nabla_h \ell(g_{\theta}(x_i + h), -1)}{\|\nabla_h \ell(g_{\theta}(x_i + h), -1)\|_2} \right).$$

Increase the negative-label loss, then project back to $r \leq \|h\|_2 \leq \gamma r$.

3. Train $x_i \mapsto +1$, $\tilde{x} \mapsto -1$; anomaly score is the negative normal-class output.

CIFAR-10: 10 one-vs-rest tasks; 5,000 normal train / 10,000 test per task.

Reproduction: 100 epochs, batch 128, no CE-only warmup; 50 ascent steps, project every 10; LR steps 40/80.

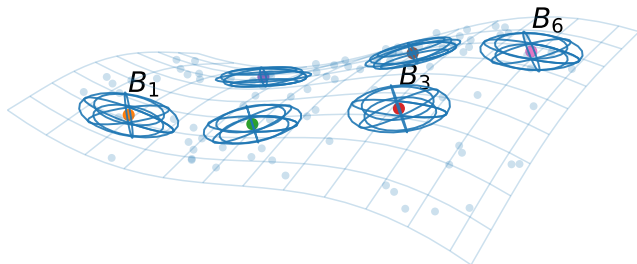
Our reproduction versus paper results

Method	Data	Paper	Ours	Δ pp
RBF SVDD ($\nu = .01$)	MNIST	91.29	95.51 ± 0.05	+4.22
	CIFAR-10	64.78	60.56 ± 0.20	-4.22
RBF SVDD ($\nu = .1$)	MNIST	91.29	95.51 ± 0.04	+4.22
	CIFAR-10	64.78	60.67 ± 0.26	-4.11
Deep SVDD: one class	MNIST	94.80	93.83 ± 0.55	-0.97
	CIFAR-10	64.81	61.70 ± 0.59	-3.11
Deep SVDD: soft	MNIST	93.53	93.30 ± 0.53	-0.23
	CIFAR-10	63.28	60.99 ± 0.48	-2.29
DROCC	CIFAR-10	74.23	65.58 ± 1.63	-8.65

Our comparison: MVTec AD

Method	AUROC	AP
PatchCore	99.18	99.77
RBFSVDD ($\nu = 0.01$)	53.09	78.18
RBFSVDD ($\nu = 0.1$)	53.21	77.91
Deep SVDD: one class	79.43	91.03
Deep SVDD: soft	76.60	89.82
DROCC	43.77	73.13
NBD: soft min + diffusion	73.56	88.01
NBD: hard min + diffusion	73.09	87.61

NBD: local geometry plus a shared graph



Each bubble stores

A center, tangent variation, and normal thickness.

The graph connects them

Diffusion compares the regions each bubble can reach.

Normal features $\rightarrow$ bubbles $\rightarrow$ shared graph $\rightarrow$ diffusion distance $\rightarrow$ query score

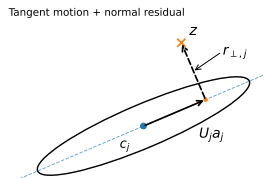
A local normality bubble

$$B_j = (c_j, U_j, \Lambda_j, \sigma_{\perp,j}^2), \quad U_j^\top U_j = I_{d_j}, \quad \Pi_j = U_j U_j^\top.$$

Local model

$$\begin{aligned} \mathbf{z} &= \mathbf{c}_j + U_j \mathbf{a} + \varepsilon_{\perp,j}, & \mathbf{a} &\perp\!\!\!\perp \varepsilon_{\perp,j}, \\ \mathbf{a} &\sim \mathcal{N}(\mathbf{0}, \Lambda_j), & \varepsilon_{\perp,j} &\sim \mathcal{N}(\mathbf{0}, \sigma_{\perp,j}^2 (\mathbf{I} - \Pi_j)). \end{aligned}$$

$$\Sigma_j = U_j \Lambda_j U_j^\top + \sigma_{\perp,j}^2 (I - \Pi_j).$$



Fit by local PCA

Neighborhood mean $\rightarrow c_j$; top d_j eigenpairs $\rightarrow U_j, \Lambda_j$.

$$\sigma_{\perp,j}^2 = \frac{1}{D - d_j} \sum_{k=d_j+1}^D \hat{\lambda}_{jk}.$$

Scoring a query

$$a_j(z) = U_j^\top (z - c_j), \quad r_{\perp,j}(z) = (I - \Pi_j)(z - c_j).$$

Local energy

$$E_j(z) = \underbrace{\frac{a_j^\top (\Lambda_j + \epsilon I)^{-1} a_j}{d_j}}_{\text{tangent rarity}} + \beta \underbrace{\frac{\|r_{\perp,j}\|_2^2}{(D - d_j)(\sigma_{\perp,j}^2 + \epsilon)}}_{\text{off-manifold residual}}.$$

Absolute fit

$$S_{\text{hard}}(z) = \min_{j \in \mathcal{C}(z)} E_j(z),$$

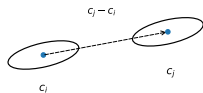
$$S_{\text{soft}}(z) = -T_b \log \left[\frac{1}{K} \sum_{j \in \mathcal{C}(z)} e^{-E_j(z)/T_b} \right].$$

Attachment to the graph

$$p_j(z) = \frac{e^{-E_j(z)/T_b}}{\sum_{k \in \mathcal{C}(z)} e^{-E_k(z)/T_b}}.$$

Connect bubbles by three local criteria

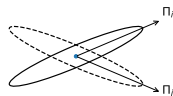
1. Center compatibility



$$r_c(i, j) = \frac{1}{2} [E_i(c_j) + E_j(c_i)].$$

Each center should be plausible under the other bubble's local geometry.

2. Tangent orientation

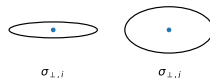


$$r_{\text{frame}}(i, j) = \|\Pi_i - \Pi_j\|_F^2$$

$$= d_i + d_j - 2 \|U_i^\top U_j\|_F^2.$$

Compare tangent subspaces, independent of basis-vector signs.

3. Normal thickness



$$r_{\perp}(i, j) = \left[\log \frac{\sigma_{\perp, i}^2 + \epsilon}{\sigma_{\perp, j}^2 + \epsilon} \right]^2.$$

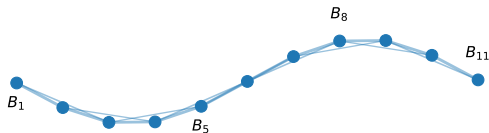
Compare residual spread, not the full covariance.

The three terms can determine graph edges. Diffusion then compares connectivity on this graph.

From bubbles to a graph

Edge dissimilarity, before diffusion

$$\delta_B(i, j) = \frac{r_c(i, j)}{\tau_c} + \lambda_{\text{frame}} \frac{r_{\text{frame}}(i, j)}{\tau_{\text{frame}}} + \lambda_{\perp} \frac{r_{\perp}(i, j)}{\tau_{\perp}}.$$



Affinity and transition

For symmetric local candidate edges $\mathcal{E}$,

$$W_{ij} = \mathbf{1}_{\{(i,j) \in \mathcal{E}\}} e^{-\delta_B(i,j)} \quad (i \neq j),$$

$$W_{ii} = \eta > 0, \quad s_i = \sum_j W_{ij},$$

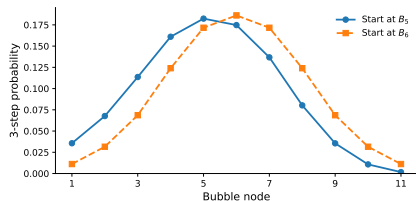
$$P_{ij} = \frac{W_{ij}}{s_i}, \quad \pi_i = \frac{s_i}{\sum_m s_m}.$$

Fit the graph once on normal data. Every query uses the same bubble nodes.

Diffusion distance

Similar t -step transition distributions give a smaller diffusion distance:

$$D_t^2(i, j) = \sum_m \frac{[(P^t)_{im} - (P^t)_{jm}]^2}{\pi_m}.$$



Compute with eigenvectors

$$P\psi_\ell = \lambda_\ell \psi_\ell, \quad \sum_i \pi_i \psi_\ell(i) \psi_k(i) = \delta_{\ell k}.$$

$$D_t^2(i, j) = \sum_{\ell=1}^{M-1} \lambda_\ell^{2t} [\psi_\ell(i) - \psi_\ell(j)]^2.$$

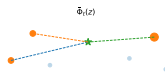
$$\Phi_t(B_j) = (\lambda_1^t \psi_1(j), \dots, \lambda_L^t \psi_L(j)),$$

$$D_{t,L}^2(i, j) = \|\Phi_t(B_i) - \Phi_t(B_j)\|^2.$$

Does the query attach to one region?



Nearby bubbles: lower dispersion



Distant bubbles: higher dispersion

$$S_D(z) = \frac{1}{2|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_{i,j} p_i(z) p_j(z) D_{t,L}^2(i,j).$$

Equivalent variance at time t

$$\bar{\Phi}_t(z) = \sum_j p_j(z) \Phi_t(B_j),$$

$$S_{D,t}(z) = \sum_j p_j(z) \left\| \Phi_t(B_j) - \bar{\Phi}_t(z) \right\|^2.$$

Why keep a local score

A sharp attachment can still have $D \approx 0$, so the local score remains necessary.

Two final scores: hard min + diffusion and soft min + diffusion

Combine local mismatch with diffusion dispersion

$$S_{\text{soft}+D}(z) = \tilde{S}_{\text{soft}}(z) + \tilde{S}_D(z), \quad S_{\text{hard}+D}(z) = \tilde{S}_{\text{hard}}(z) + \tilde{S}_D(z).$$

Put the components on a common scale

For each category, compare a score to its distribution on reserved normal patches:

$$\tilde{S}_q(z) = -\log \left[\frac{1 + \sum_{u \in \mathcal{Z}_{\text{cal}}} \mathbf{1}_{\{S_q(u) \geq S_q(z)\}}}{|\mathcal{Z}_{\text{cal}}| + 1} \right], \quad q \in \{\text{soft}, \text{hard}, D\}.$$

From patches to an image

Average the 8 largest combined patch scores, using the same image aggregation as the feature-space controls.

NBD controls: adapted models on shared features

Method	Image AUROC (%)	AP (%)
PatchScore: same centers	95.86 ± 0.23	98.67 ± 0.05
PatchScore: matched memory	98.57 ± 0.21	99.59 ± 0.06
RBFSVDD (feature model)	78.20 ± 0.26	90.46 ± 0.17
Deep SVDD (feature head)	81.15 ± 0.18	91.24 ± 0.13
DROCC (feature head)	57.24 ± 3.52	76.95 ± 1.83
NBD: soft min + diffusion	76.16 ± 2.72	89.55 ± 1.60
NBD: hard min + diffusion	75.64 ± 2.64	89.18 ± 1.59

All baselines by MVTec category

Category	PatchScore center	PatchScore memory	RBF SVDD	Deep SVDD	DROCC	NBD soft + D
bottle	100.00	100.00	77.04	97.83	58.25	46.67
cable	97.20	99.89	78.18	70.09	47.14	74.22
capsule	92.75	97.07	60.44	74.11	51.29	67.40
carpet	98.86	98.66	98.62	90.68	77.49	85.39
grid	93.73	96.07	60.65	68.28	39.39	73.91
hazelnut	100.00	100.00	87.74	72.26	43.73	64.94
leather	100.00	100.00	100.00	99.78	87.22	89.65
metal nut	97.54	99.82	34.82	87.41	51.55	59.17
pill	87.50	94.72	67.47	76.14	48.37	64.97
screw	78.75	95.31	57.45	48.04	56.78	68.81
tile	98.87	98.68	99.00	87.61	76.64	99.42
toothbrush	95.93	99.72	86.48	79.26	52.41	81.11
transistor	99.29	99.96	77.56	79.04	51.26	92.89
wood	99.01	99.18	98.25	96.17	50.56	77.66
zipper	98.44	99.40	89.38	90.55	66.55	96.19

Does adding diffusion improve the local score?

Local score	Local AUROC (%)	With diffusion (%)	Δ pp
soft min	82.47 ± 1.75	76.16 ± 2.72	-6.31
hard min	82.71 ± 1.81	75.64 ± 2.64	-7.06

Overall, adding D reduces AUROC

Soft-min route: -6.31 percentage points.

Hard-min route: -7.06 percentage points.

Where does diffusion help or hurt?

Category	soft min + diffusion vs. soft min (pp)	hard min + diffusion vs. hard min (pp)
bottle	-7.67 ± 1.60	-8.02 ± 0.92
cable	-12.49 ± 2.59	-13.77 ± 3.00
capsule	-10.04 ± 3.62	-10.53 ± 3.33
carpet	-1.99 ± 4.57	-3.09 ± 4.67
grid	-13.92 ± 0.60	-14.17 ± 0.75
hazelnut	-9.71 ± 2.03	-9.54 ± 1.09
leather	-5.14 ± 1.99	-7.78 ± 5.43
metal_nut	-4.43 ± 2.74	-4.77 ± 2.13
pill	$+2.37 \pm 3.00$	$+1.45 \pm 2.00$
screw	-4.00 ± 1.19	-4.09 ± 1.59
tile	$+0.47 \pm 0.28$	$+0.63 \pm 0.40$
toothbrush	-3.89 ± 1.92	-4.26 ± 2.43
transistor	-4.13 ± 1.27	-5.76 ± 1.44
wood	-18.33 ± 16.14	-20.13 ± 16.57
zipper	-1.69 ± 0.98	-2.12 ± 1.66

What the experiment shows

1. Frozen features set a high bar

The ImageNet-pretrained Wide ResNet descriptor likely already captures much nonlinear normal structure: PatchScore with a normal memory bank reaches **98.57% AUROC**.

2. Finding: compact centers retain most of PatchScore

Replacing the **2,262**-vector PatchScore memory bank with only **128** bubble centers ($\sim 17.7\times$ **fewer representatives**) still reaches **95.86% AUROC**, just **2.71 pp** lower. The bubble construction therefore acts as a compact normal memory while preserving most of the useful structure.

3. The current limitation is diffusion

Soft-min + diffusion loses **6.31 pp**; hard-min + diffusion loses **7.06 pp**. On these strong features, the evidence favors the local center score; next isolate graph connectivity and component scaling.

Appendix: feature-head training diagnostics

Method	Diagnostic across 45 runs	Value
Deep SVDD head	Median output-variance ratio	0.0008381
	Ratio range	0.0001495–0.0181573
DROCC head	Median final training loss	1.3862944
	Mean test true-positive rate	14.07%

Interpretation

The Deep SVDD feature head strongly contracts normal outputs. Variance alone cannot establish a constant mapping.

The DROCC feature head's weak test detection accompanies a high final training loss in this feature-space setting.

Method references

- [1] Tax and Duin (2004).
Support Vector Data Description.
- [2] Ruff et al. (2018).
Deep One-Class Classification.
- [3] Goyal et al. (2020).
DROCC.
- [4] Coifman and Lafon (2006).
Diffusion maps.
- [5] Roth et al. (2022).
PatchCore paper. Author implementation.

Experiment materials

[Our baseline benchmark results](#)

[NBD study: category results](#)

[Fixed experiment configuration](#)

[Benchmark implementation](#)

MVTec AD dataset